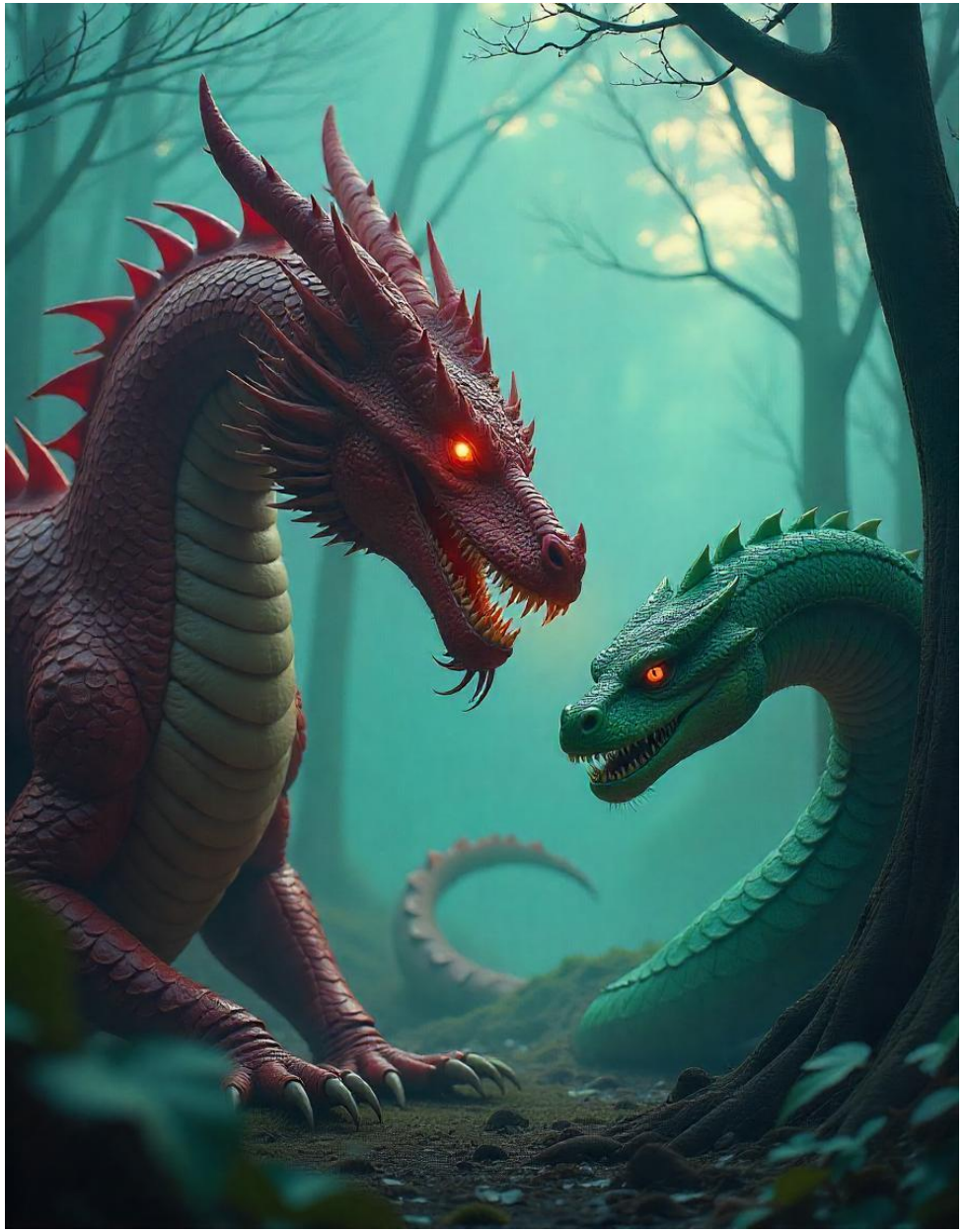
Francesco Conte

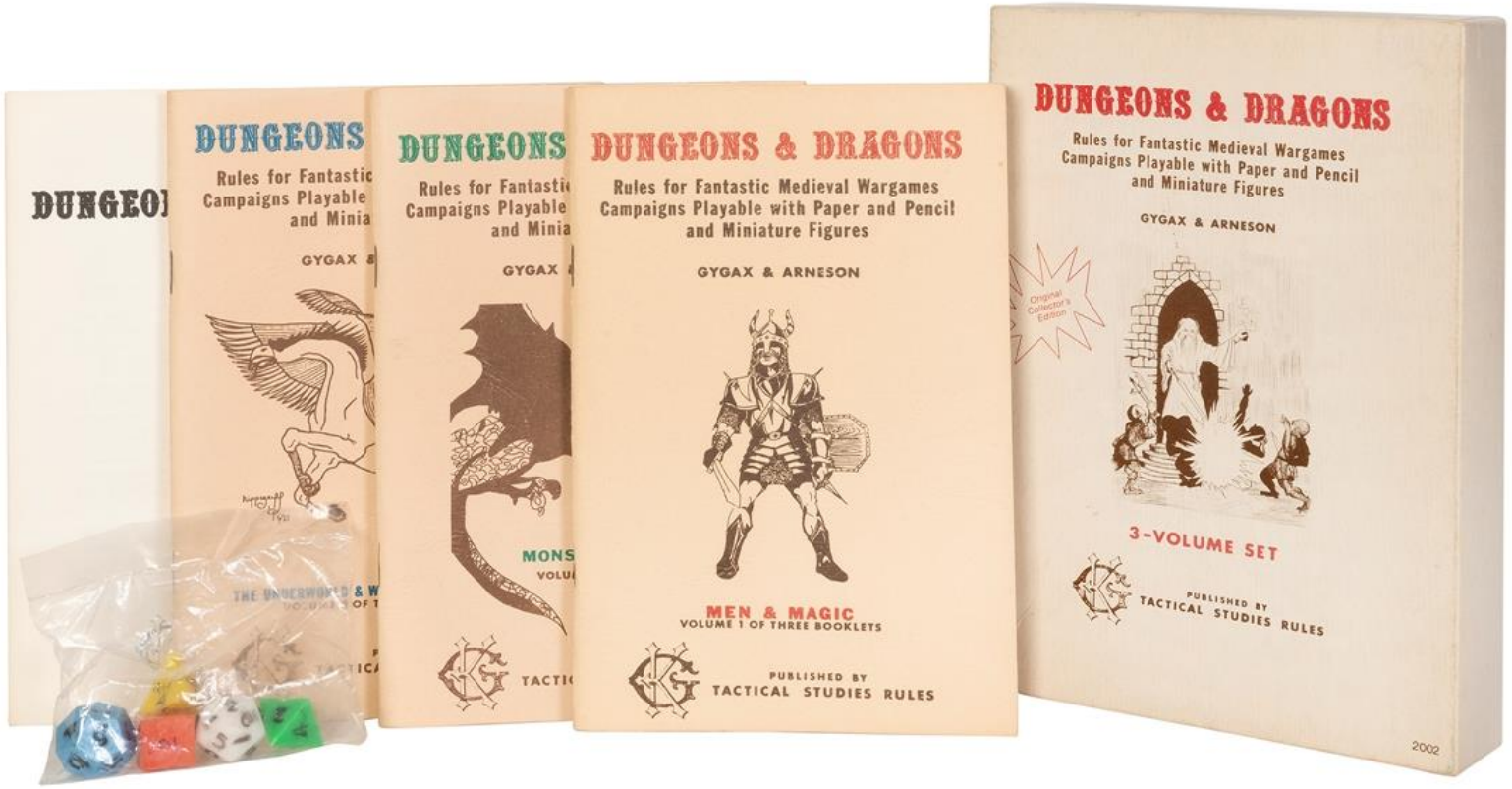
DungeonPy

Exploiting Python tools for your hobbies
(when you probably shouldn’t)



Dungeons & Dragons

- …1973!
- Most famous fantasy rpg
- Great for networking, making friendships…
- One **DM** (Dungeon Master)
- Many **Players** and **PCs** (Player Characters)



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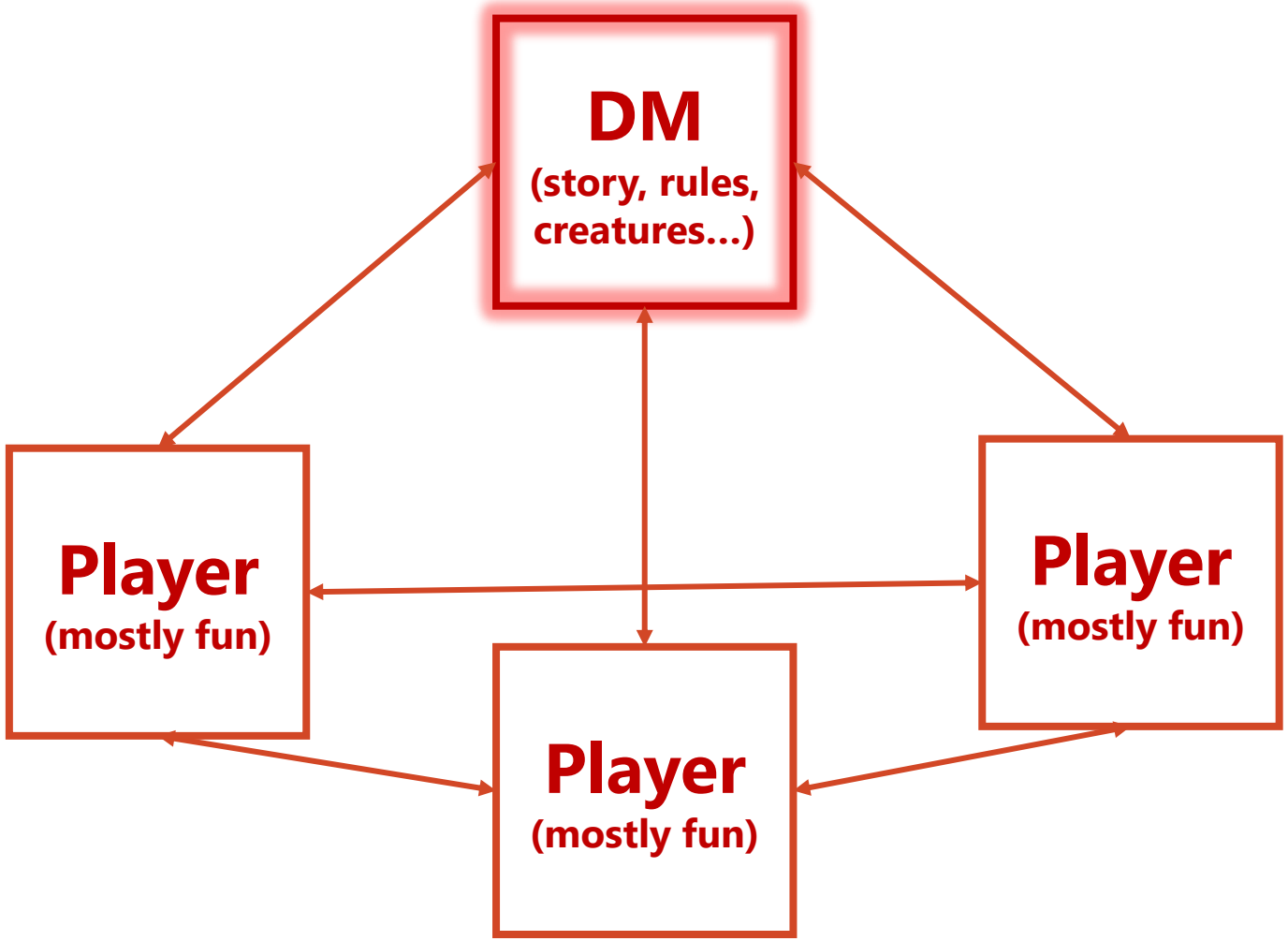

Dungeons & Dragons in 2020

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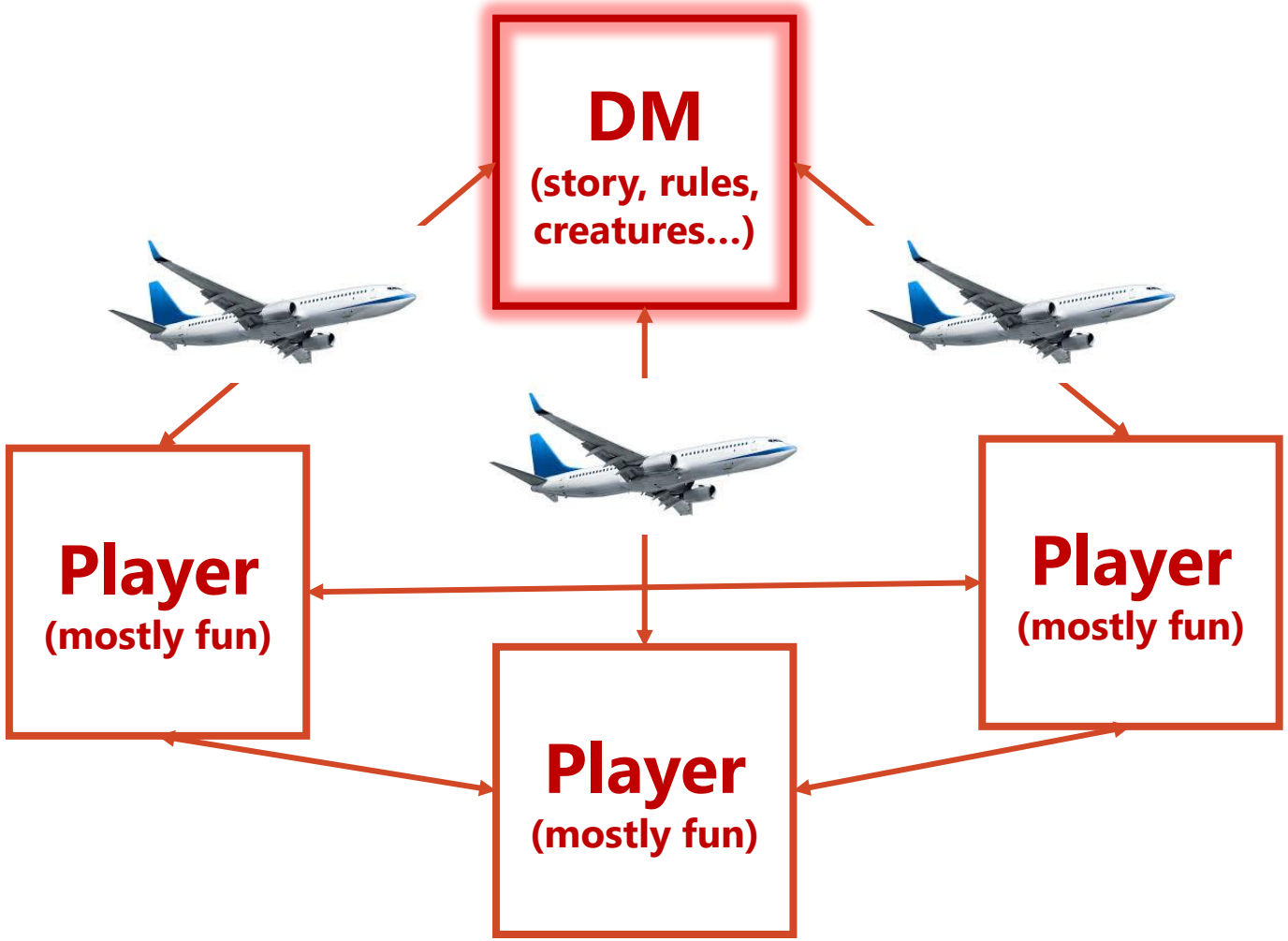
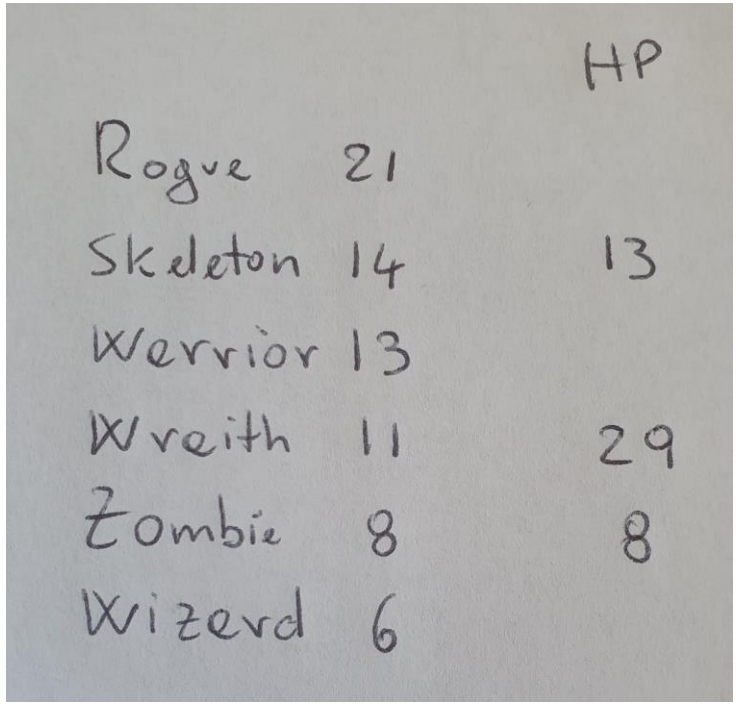


A simple D&D GUI for the Initiative Tracker

Initiative tracker:

- Helps immensely with the round organization
- Tracks turn order, conditions, creature health…



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A simple D&D GUI for the Initiative Tracker

```
import PySimpleGUI as sg
layout = [
    [sg.Text('Initiative Tracker', font=('Helvetica', font_size))],
    [sg.Table(values=[], headings=['Name', 'Initiative', 'HP', 'Conditions'],
        auto_size_columns=False, justification='left', col_widths=[20, 10, 10, 20],
        key='-TABLE-', enable_events=True, row_height=25, expand_x=True, num_rows=10,
        background_color='white', text_color='black', font=('Segoe UI Emoji', font_size))],
    [sg.Text('Turn:', font=('Segoe UI Emoji', font_size)), sg.Input('1', key='-TURN-', size=(5, 1), font=('Segoe UI Emoji',
     sg.Button('⏮ Prev Char', font=('Segoe UI Emoji', font_size)), sg.Button('⏭ Next Char', font=('Segoe UI Emoji', font_si
    [sg.HorizontalSeparator()],
    [sg.Text('Name', size=(10, 1), font=('Segoe UI Emoji', font_size)), sg.Input(key='-NAME-', size=(30, 1), font=('Segoe UI
    [sg.Text('Initiative', size=(10, 1), font=('Segoe UI Emoji', font_size)), sg.Input(key='-INITIATIVE-', size=(5, 1), font
    [sg.Text('HP (optional):', size=(12, 1), font=('Segoe UI Emoji', font_size)), sg.Input(key='-HP-', size=(5, 1), font=('S
     sg.Text('   Change:', font=('Segoe UI Emoji', font_size)), sg.Input('0', key='-HP_CHANGE-', size=(5, 1), font=('Segoe U
     sg.Button('Wound', font=('Segoe UI Emoji', font_size)), sg.Button('Heal', font=('Segoe UI Emoji', font_size))],
    [sg.Text('Conditions:', font=('Segoe UI Emoji', font_size))],
    *condition_rows,
    [sg.Button('Add New', font=('Segoe UI Emoji', font_size)), sg.Button('Update Selected', font=('Segoe UI Emoji', font_siz
        sg.Button('Delete Selected', font=('Segoe UI Emoji', font_size)), sg.Button('↑ Move Up', font=('Segoe UI Emoji', fon
        sg.Button('↓ Move Down', font=('Segoe UI Emoji', font_size))],
    [sg.Button('💾 Export', font=('Segoe UI Emoji', font_size)), sg.Button('📂 Load', font=('Segoe UI Emoji', font_size))]

]
window = sg.Window('D&D Initiative Tracker', layout, resizable=True)
```

PySimpleGUI

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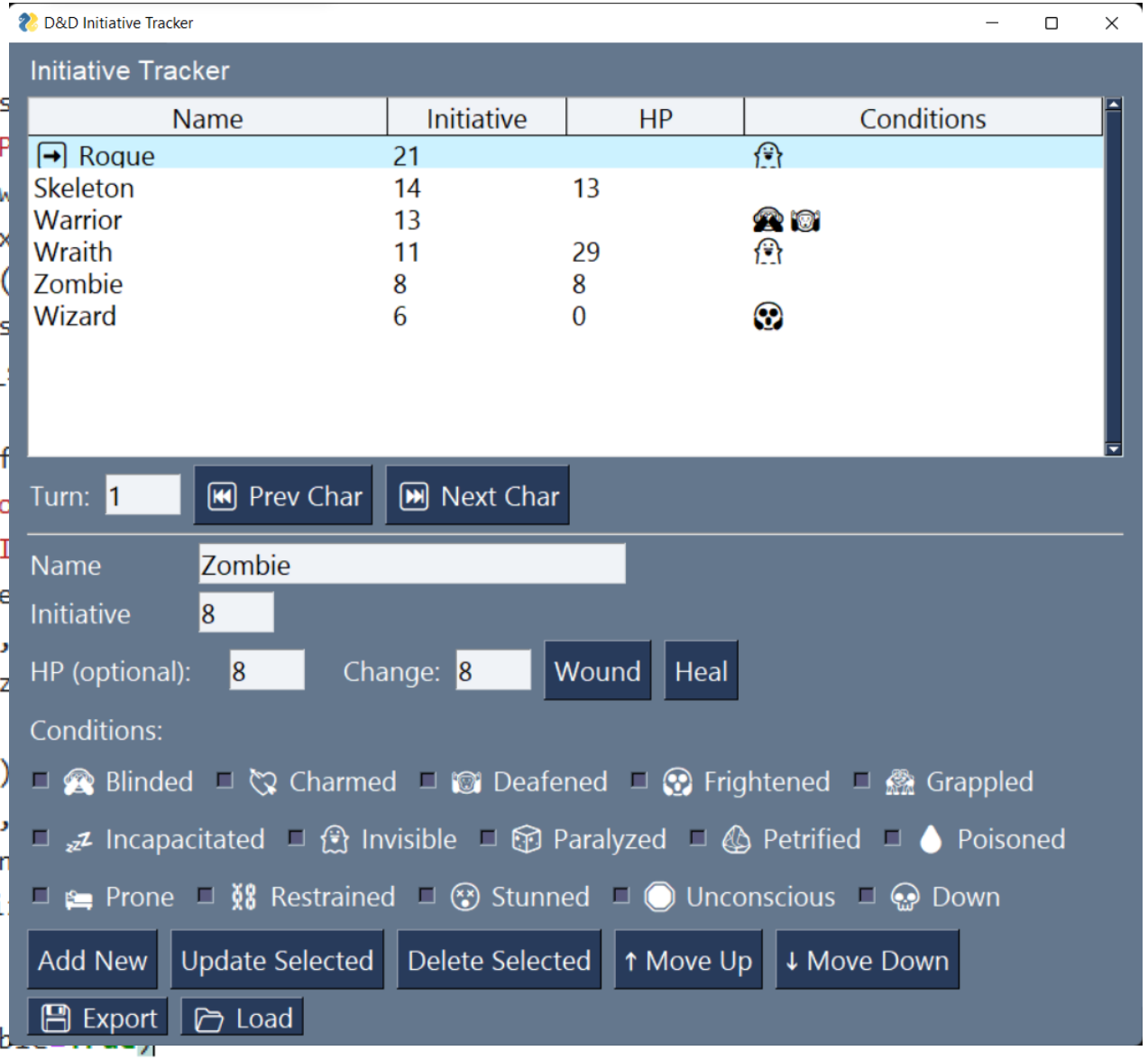

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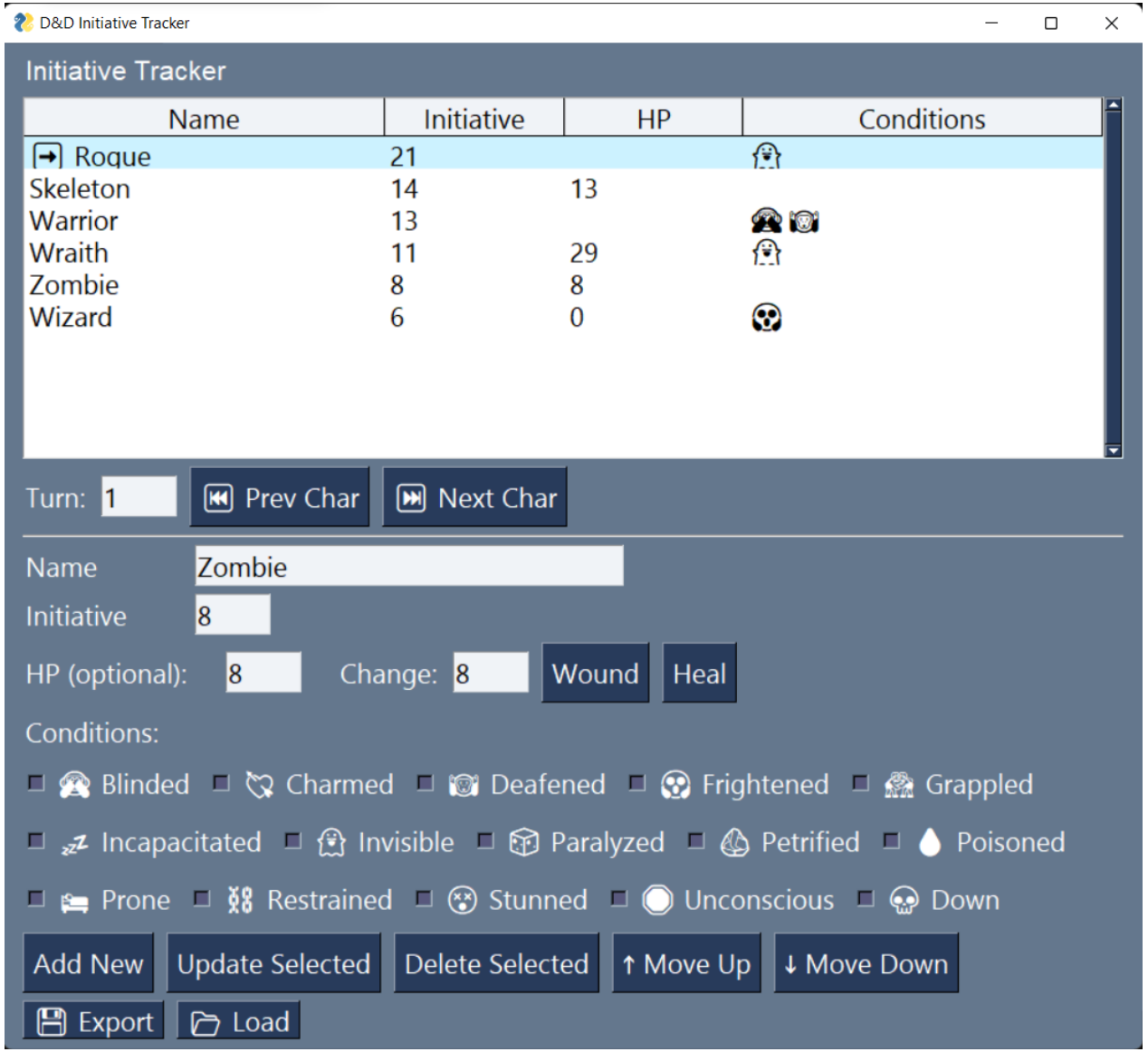

A simple D&D GUI for the Initiative Tracker

Initiative tracker:

- Helps immensely with the round organization
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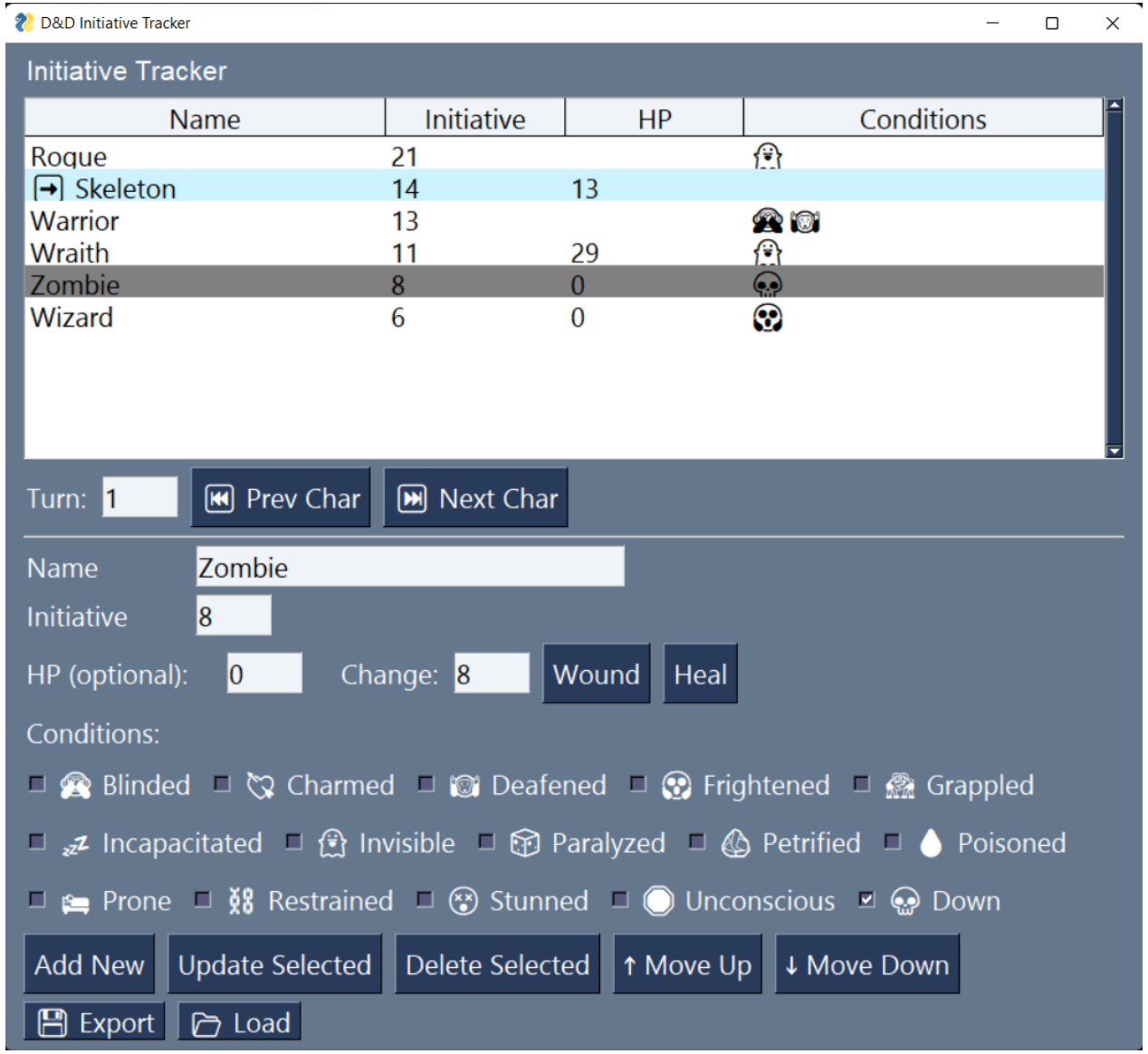
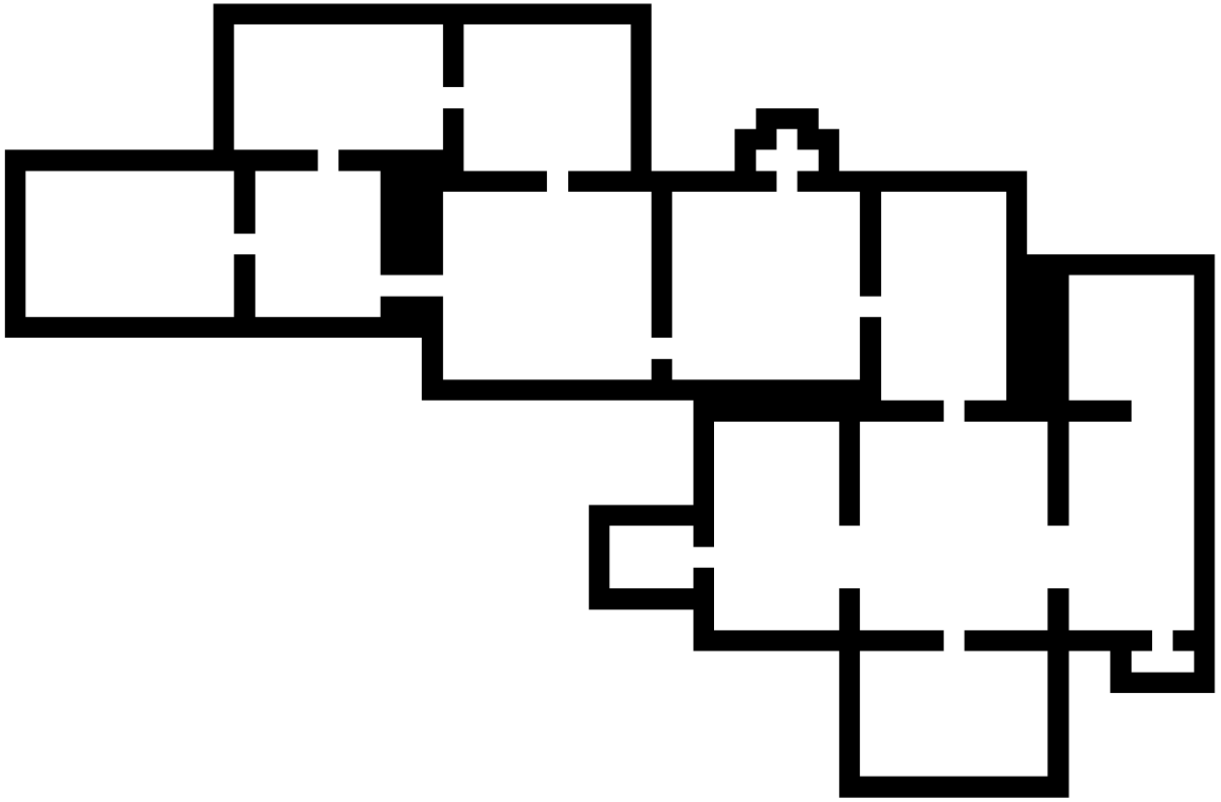
Since I have a tracker, why not sharing a map on the Zoom screen?



A simple D&D map to share on-screen (...)



A simple D&D map to share on-screen (…)

```
2222222222222222222222222222222222222222222222222222222222
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2222222222210000000001000000001222211011222222222222222222
2111111111111111311111100000000122221000122222222222222222
2100000000001000000111111113111111111141111111111222222222
2100000000001000000111000000000010000000010000001222222222
2100000000001000000111000000000010000000010000001222222222
2100000000003000000111000000000010000000010000001222222222
2100000000001000000111000000000010000000010000001111111111
2100000000001000000000000000000010000000010000001110000001
2100000000001000000111000000000010000000030000001110000001
2111111111111111111111000000000010000000010000001110000001
2222222222222222222221000000000030000000010000001110000001
2222222222222222222221000000000010000000010000001110000001
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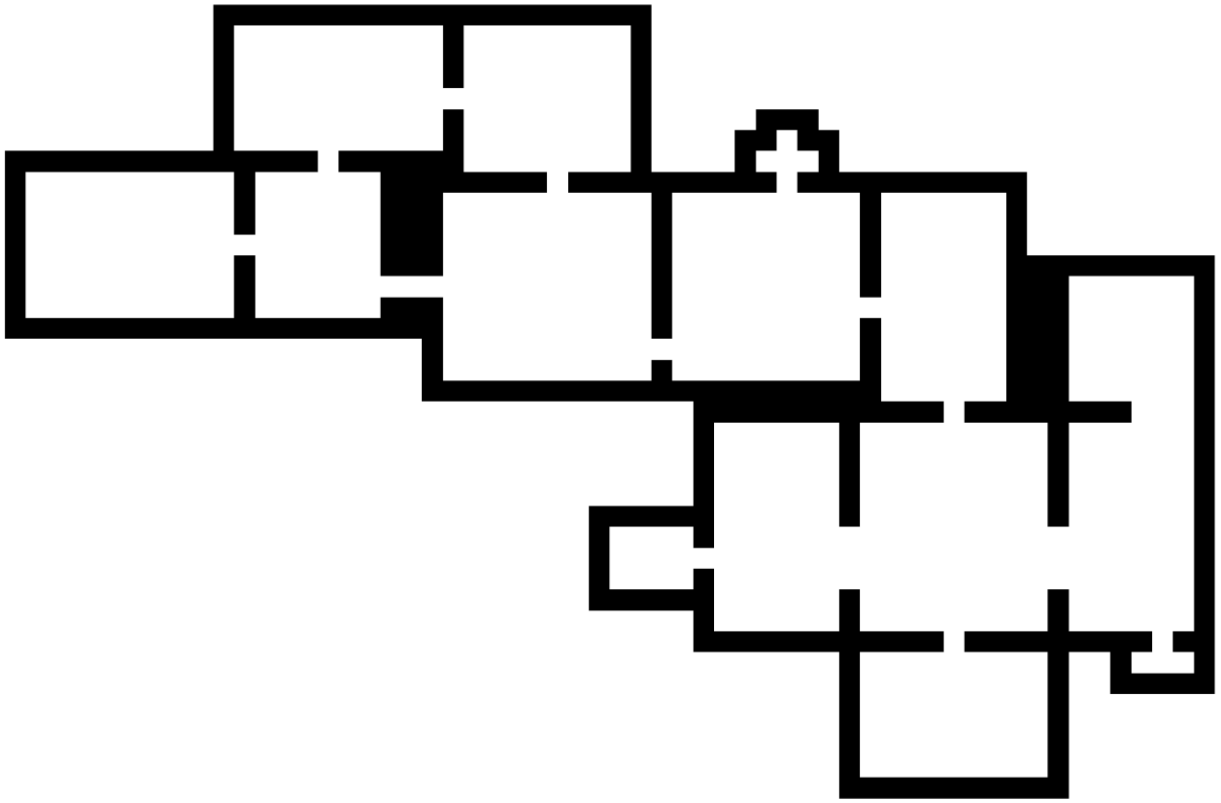

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A simple D&D map to share on-screen (...)

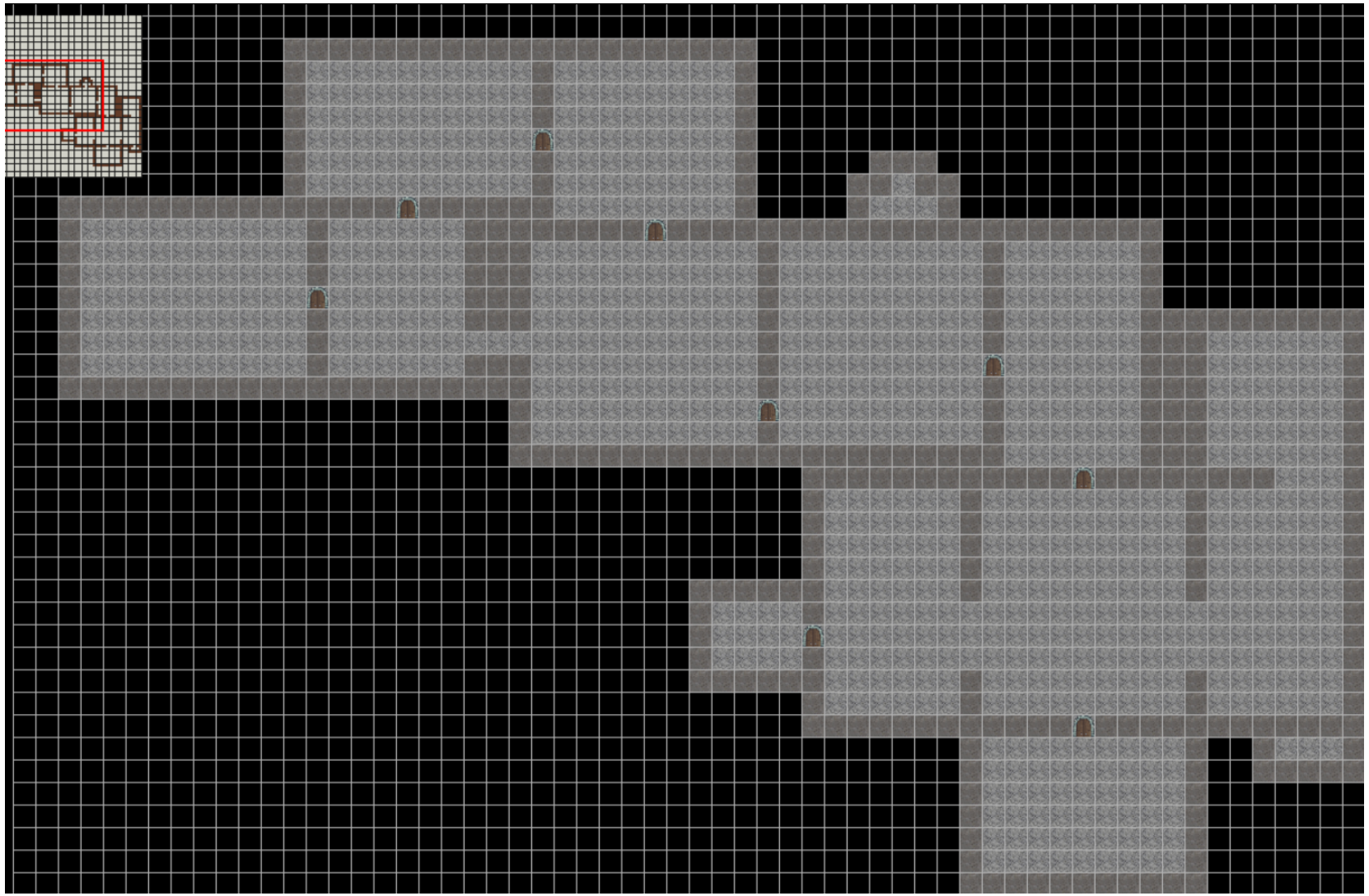
```python
import pygame
# === PYGAME INIT ===
pygame.init()
info = pygame.display.Info()
SCREEN_WIDTH, SCREEN_HEIGHT = int(info.current_w * 0.5), int(info.current_h * 0.5)
fullscreen = False
screen = pygame.display.set_mode((SCREEN_WIDTH, SCREEN_HEIGHT), pygame.RESIZABLE)
pygame.display.set_caption("D&D Map Grid")
clock = pygame.time.Clock()

# === TEXTURES ===
WALL_COLOR = (100, 60, 40)
FLOOR_COLOR = (210, 210, 200)
floor_texture_original = pygame.image.load(dir_path + 'stonefloor3.jpg').convert()
wall_texture_original = pygame.image.load(dir_path + 'stonefloor4.jpg').convert()
door_texture_original = pygame.image.load(dir_path + 'closed_door1.png').convert_alpha()
sdoor_texture_original = wall_texture_original

def scale_textures(tile_size):
    return (
        pygame.transform.scale(floor_texture_original, (tile_size, tile_size)),
        pygame.transform.scale(wall_texture_original, (tile_size, tile_size)),
        pygame.transform.scale(door_texture_original, (tile_size, tile_size)),
        pygame.transform.scale(sdoor_texture_original, (tile_size, tile_size))
    )
floor_texture, wall_texture, door_texture, sdoor_texture = scale_textures(TILE_SIZE)
```

pygame

A simple D&D map to share on-screen (…)

Interactive map

- Can zoom
- Can drag&drop icons
- Interactive minimap
- …



A simple D&D map to share on-screen (…)

Interactive map

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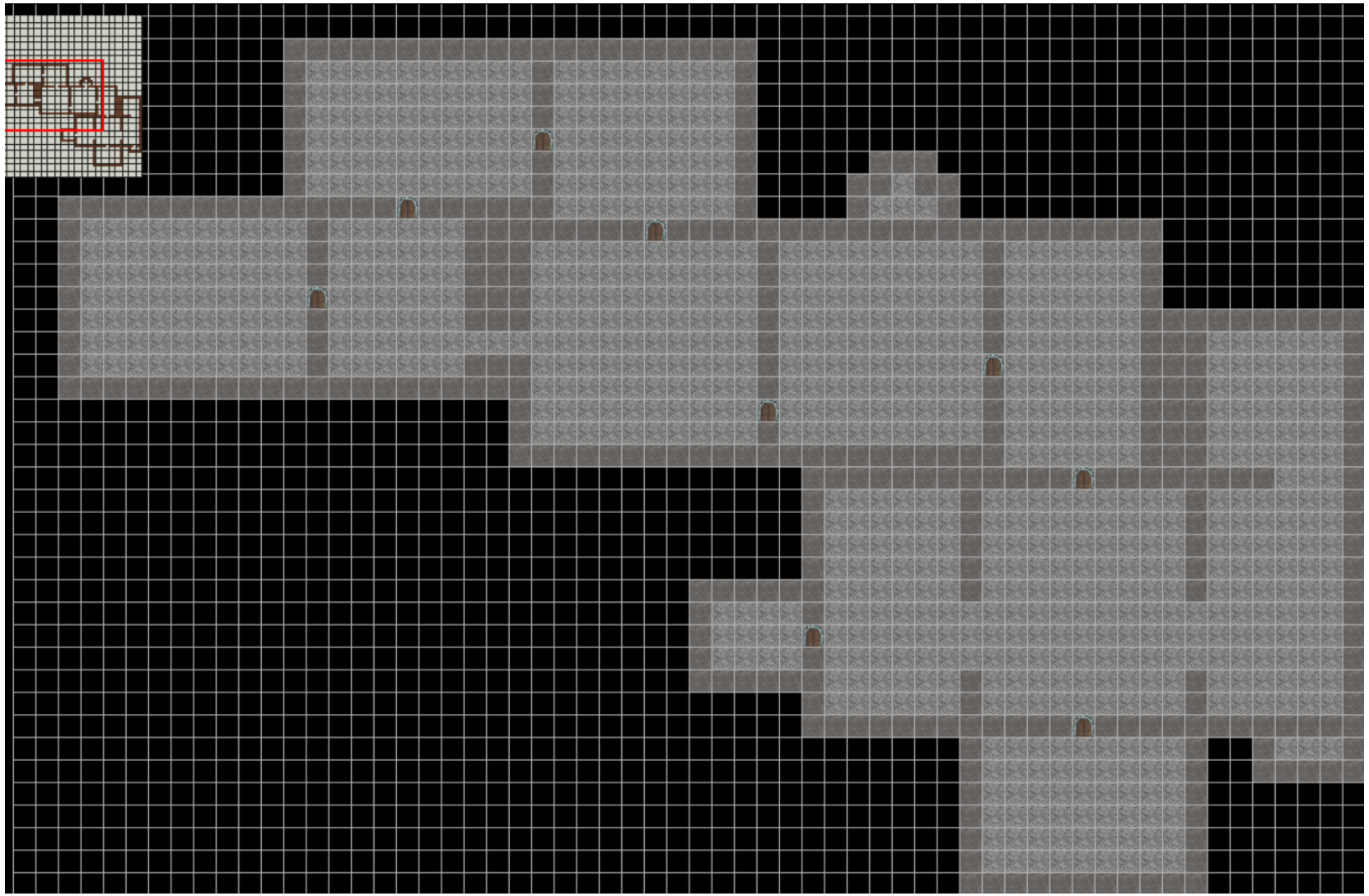


I really should have stopped there

Interactive map

- Can zoom
- Can drag&drop icons
- Interactive minimap
- …

Why don't I sync the map to the initiative tracker?

- Could load the creatures with the position
- Could link creatures' stats with their positions
- Could interact interchangeably between the two windows
- Could do many exciting things!

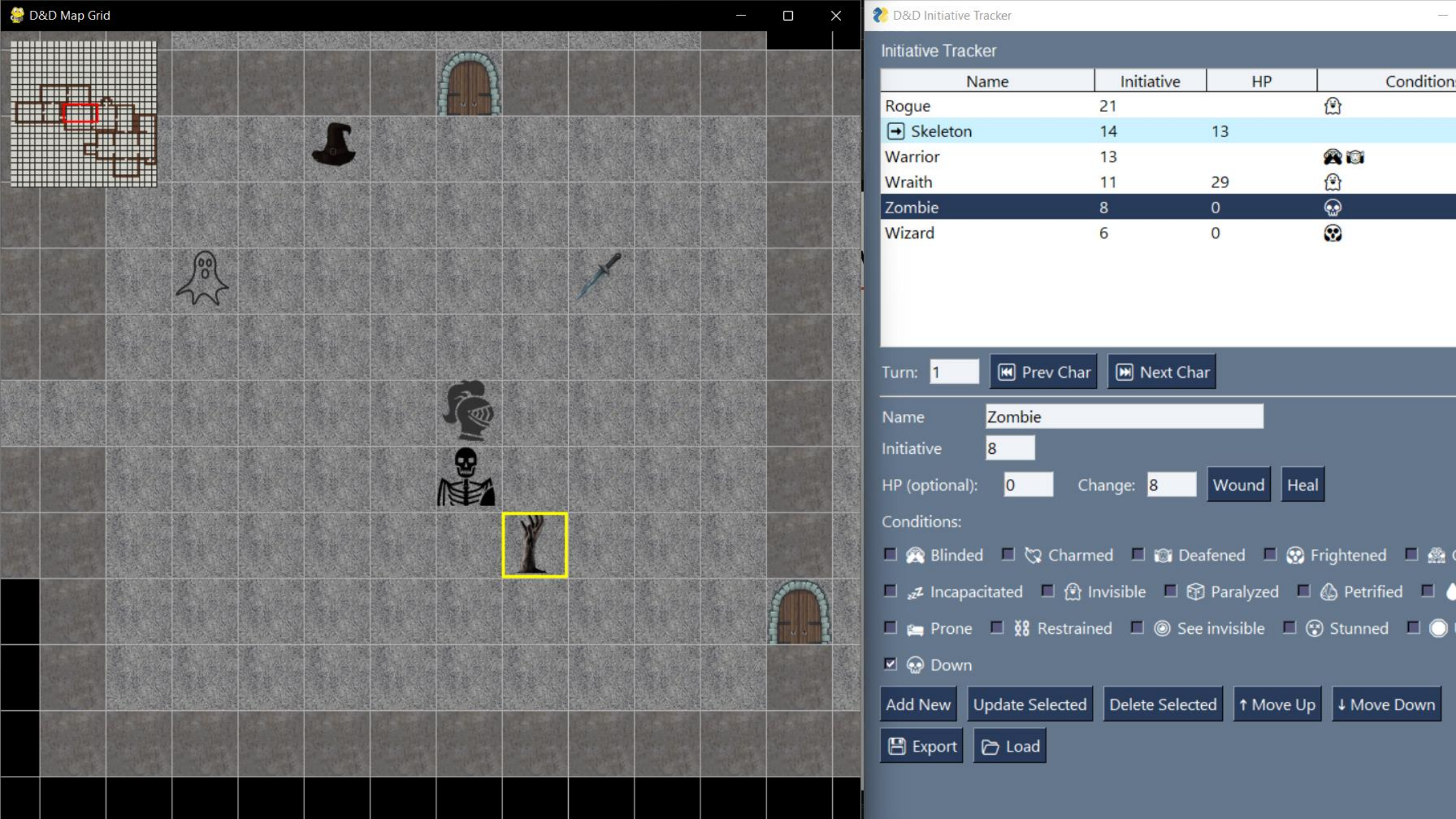
Setting up a listener

```python
selected_combatant_file = dir_path + "selected_combatant.json"

def check_external_selection():
    if os.path.exists(selected_combatant_file):
        try:
            with open(selected_combatant_file, "r", encoding="utf-8") as f:
                data = json.load(f)
            target_name = data.get("selected")
            for idx, entry in enumerate(initiative_data):
                if entry["name"] == target_name:
                    window['-TABLE-'].update(select_rows=[idx])
                    break
        except Exception as e:
            print(f"Error reading external selection: {e}")

while True:
    event, values = window.read(timeout=200)
    check_external_selection()
```

(Tracker side)

(Map side)

```python
def set_selected_combatant(name):
    with open(dir_path + "selected_combatant.json", "w", encoding="utf-8") as f:
        json.dump({"selected": name}, f)
```


D&D Map Grid
D&D Initiative Tracker
Initiative Tracker
Name
Initiative
HP
Conditions
Rogue 21
Skeleton 14 13
Warrior 13
Wraith 11 29
Zombie 8 0
Wizard 6 0
Turn: 1
Prev Char
Next Char
Name Zombie
Initiative 8
HP (optional): 0
Change: 8
Wound
Heal
Conditions:
Blinded
Charmed
Deafened
Frightened
Incapacitated
Invisible
Paralyzed
Petrified
Prone
Restrained
See invisible
Stunned
Down
Add New
Update Selected
Delete Selected
↑ Move Up
↓ Move Down
Export
Load

I REALLY should have stopped there

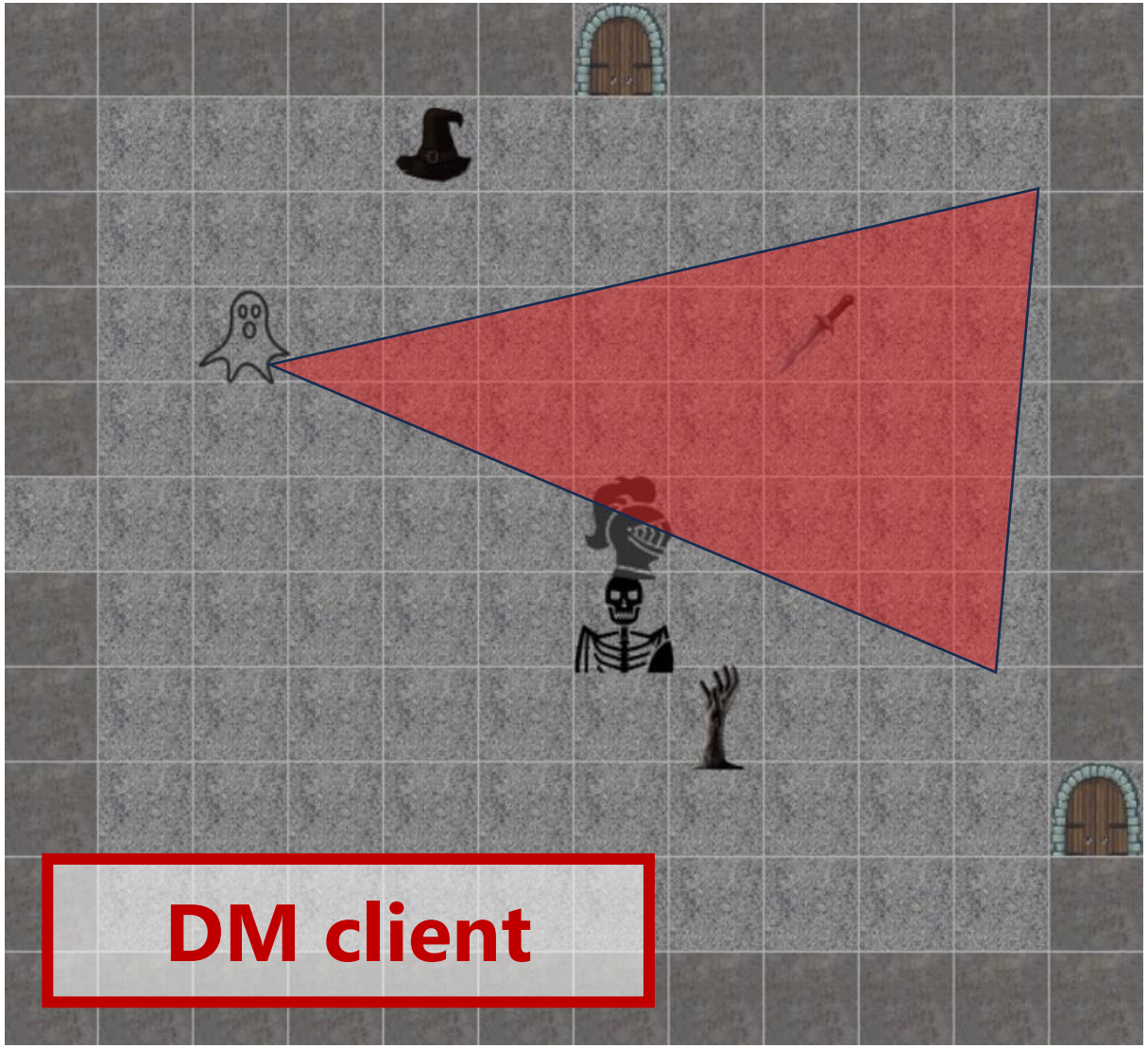
Integrated map and tracker

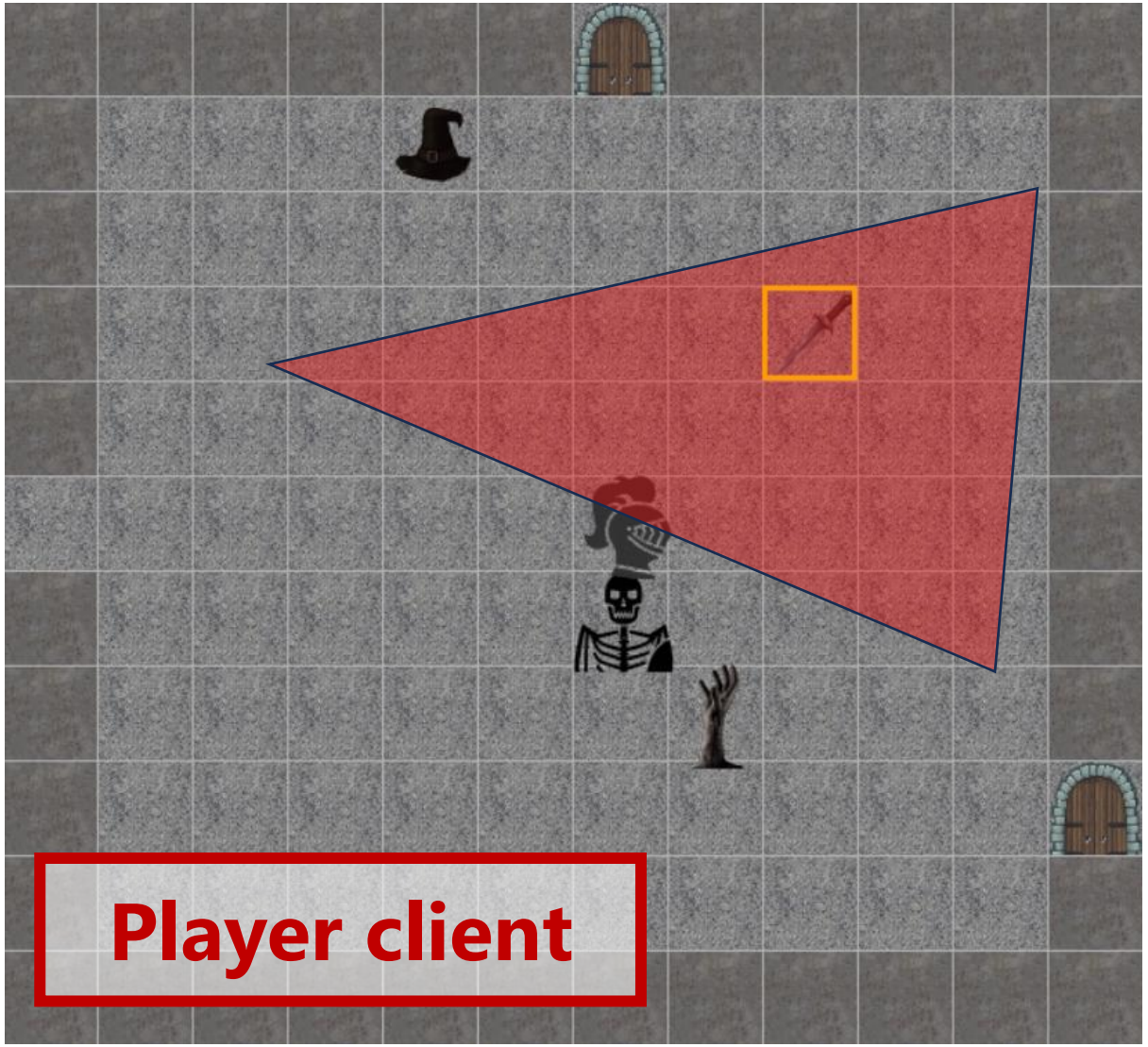
- Can select creatures, and their stars
- Can drag & drop, update positions, handle visually dynamic strategies and tactics
- Can overlay area effects like circular explosions and cones
- Thousands of possibilities!

What if I set up a server?

- The players could have their own client windows
- The DM (me) sees the tracker and the full map
- The players visualize only
- All of them could interact the part of the map that their characters see or know on the screen, overlaying effects, moving their icon, etc (needed to enforce stronger behavioral rules to avoid mayhem)
- Could do many exciting things!

...or maybe not?





Thank you for listening!

WE DON'T STOP PLAYING BECAUSE WE GROW OLD;
WE GROW OLD BECAUSE WE STOP PLAYING

(G.B.Shaw)